Skills Summary

Evaluating Piecewise and Absolute-Value Functions

Use this reference while completing your worksheet or when studying.

Skill 1 — Absolute value is a distance

Meaning: $|a|$ is how far a is from zero, so $|-7| = 7$ and $|7| = 7$. It is never negative.

Order: finish everything *inside* the bars, take the distance, then do what is outside. A minus sign in front of the bars is outside, so it acts last — and it can make the whole value negative.

Example: For $f(x) = |x - 5| + 1$, find $f(2)$.
Step 1. This is a substitution, so the only decision is the order — inside the bars first, because the bars group like brackets.
Step 2. $2 - 5 = -3$; the distance of -3 from zero is 3; then add the 1 outside.
$f(2) = \mathbf{4}$

Try it: For $g(x) = -3|x + 2| + 10$, find $g(1)$.

check: $g(1) = \mathbf{1}$

(!) Watch out: $-3|x + 2|$ does *not* mean $|-3(x + 2)|$. The bars close before the -3 is applied, so the result of the bars is positive and the -3 then makes it negative.

Skill 2 — Pick the rule before you compute

A piecewise function is several rules, each with a condition. **Test the input against the conditions first**, then use only the rule it satisfies.

$<$ excludes the boundary and $\leq$ includes it, so exactly one rule ever owns a boundary value — and it is the one written with $\leq$ or $\geq$.

Example: For $f(x) = 2x + 3$ when $x < 1$, and $x^2 + 1$ when $x \geq 1$, find $f(0)$ and $f(3)$.
Step 1. Nothing can be computed until each input is matched to a condition, so read the conditions before looking at the rules.
Step 2. $0 < 1$, so use $2x + 3$: $2(0) + 3$. And $3 \geq 1$, so use $x^2 + 1$: $3^2 + 1$.
$f(0) = \mathbf{3}$, $f(3) = \mathbf{10}$

Try it: For the same f, find $f(1)$.

check: $f(1) = \mathbf{2}$

Skill 3 — Working backwards to the input

Given the output, find the inputs. Undo the operations *outside* the bars first, until the absolute value stands alone; then split into two equations, because two numbers sit at the same distance from zero.

$|A| = k$ (with $k > 0$) becomes $A = k$ *or* $A = -k$. Solving only the first loses half the answer.

Example: Solve $|x - 4| = 6$.
Step 1. The bars are already alone, so the job is the split — distance 6 can be reached from either side, so expect two answers.
Step 2. $x - 4 = 6$ gives $x = 10$; $x - 4 = -6$ gives $x = -2$.
$x = \mathbf{-2}$ or $x = \mathbf{10}$

Try it: Solve $2|x + 3| = 8$.

check: $x = \mathbf{-7}$ or $x = \mathbf{1}$